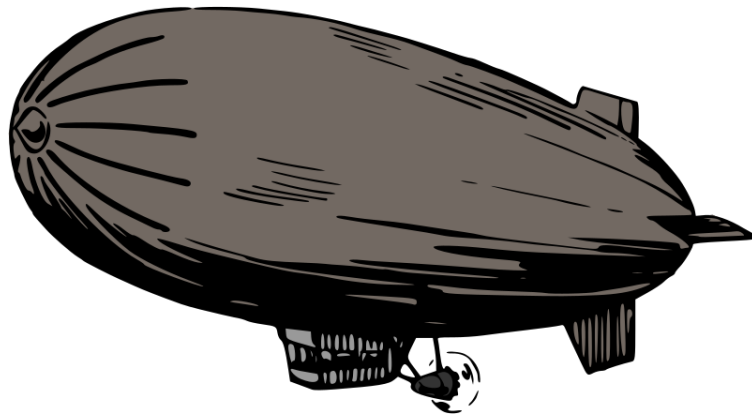


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**SYSTEM REQUIREMENTS SPECIFICATION
CSE 4316: SENIOR DESIGN I
SUMMER 2025**



**AIRSHIP CREW
PROJECT AIRSHIP**

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1 PRODUCT CONCEPT

This section describes the purpose, use, and intended user audience for the Autonomous Cargo Blimp system. The Autonomous Cargo Blimp is an eco-friendly aerial logistics platform designed to autonomously transport cargo over short to medium distances without the need for traditional infrastructure. The system is equipped with GPS navigation, helium-based lift, electric propulsion, and safety protocols for reliable flight and landing. This section provides a high-level overview of what the system is designed to do and who it is intended to serve.

1.1 PURPOSE AND USE

The Autonomous Cargo Blimp is designed to be a low-emission, self-navigating aerial system designed to transport lightweight cargo across local areas. It uses helium for lift and electric motors for propulsion and directional control, enabling vertical takeoff and landing (VTOL). GPS and sensor input allow it to follow predefined routes, maintain stable altitude using a PID control system, and avoid obstacles during flight. Communication between the blimp and a ground station is handled via Wi-Fi, with emergency protocols in place to ensure safe landings if signals are lost or battery levels are low. While cargo handling is a future goal, the focus is on achieving stable autonomous flight and navigation. This product aligns with emerging trends in sustainable transportation technologies [7].

1.2 INTENDED AUDIENCE

The intended audience for the Autonomous Cargo Blimp includes logistics providers seeking environmentally sustainable transport solutions, humanitarian organizations aiming to deliver supplies to disaster-affected or infrastructure-limited areas, and academic researchers or commercial developers working in autonomous systems and aerial robotics. Although this system is currently being developed as a student prototype, it serves as a model for scalable future systems and could eventually be integrated into larger, more complex logistics infrastructures.

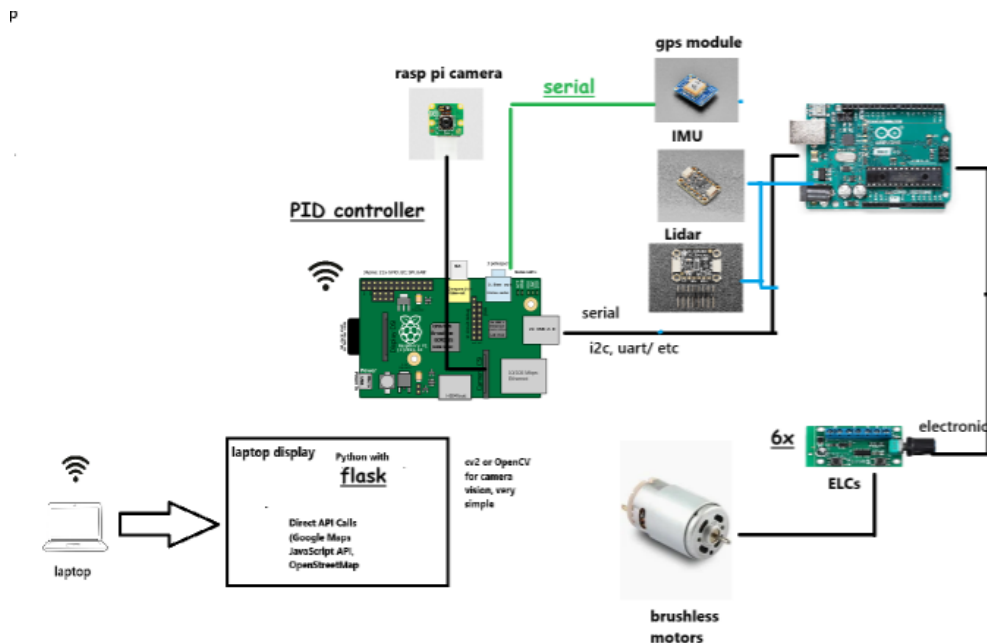


Figure 1: System Conceptual Diagram

2 PRODUCT DESCRIPTION

This section provides the reader with an overview of Autonomous Cargo Airship. The primary operational aspects of the product, from the perspective of end users, maintainers and administrators, are defined here. The key features and functions found in the product, as well as critical user interactions and user interfaces are described in detail.

2.1 FEATURES & FUNCTIONS

The autonomous airship will be able to navigate at a specified altitude while carrying a load within its gondola. It will have the ability to fly through the air using both autonomous navigation and manual control from its ground station. Additionally, the blimp will navigate around obstacles on its route to ensure a safe arrival at its destination. In Figure 1, the diagram illustrates how the hardware and software will communicate with each other to achieve the specified objectives. It should be noted that the blimp will not unload and load the delivery items itself; instead, it will require human intervention.

Description of components in Figure 1:

- **Laptop/Ground Station:** Provides a user interface, live system readings, waypoint management, and communication with the Raspberry Pi.
- **Raspberry Pi:** Uses wireless communication with the ground station, communicates with the Arduino to send motor commands and receive sensor data, receives GPS data, and streams video.
- **Arduino:** Reads sensor data to deliver to Raspberry Pi, directs motor controls from Raspberry Pi, and operates a fail-safe.

The required external elements include a GPS satellite and a Wi-Fi network.

2.2 EXTERNAL INPUTS & OUTPUTS

Data Element	Description	Use
Ground Station Flight Commands	Enables manual control over the blimp allowing the user to navigate the blimp using remote controls.	Provides a layer of stability and control if autonomous mode is not enabled.
GPS Coordinates	The airship's coordinates are calculated from a satellite and transferred to the GPS module.	Facilitates auto-navigation by specifying a destination and reaching it from its current coordinates. It will also inform the Ground Station of where the airship is located.
System Status	Displays live readings of the blimp's sensors data, battery level, and telemetry data on the Ground Station	Renders live diagnostics of the blimp's status and current operation.
Airship's Vision	Streams the camera's feed	Transmits the airship's environment for the user and obstacle avoidance

2.3 PRODUCT INTERFACES

System Status - Sensor Data:

```
Inertial Measurement Unit Data
      Temperature 27.23 deg C
      Accel X: 7.68   Y: -5.55       Z: 2.74 m/s^2
      Mag X: -52.50  Y: -31.50      Z: -0.30 uT
      Gyro X: 0.01   Y: -0.00      Z: -0.00 radians/s
```

Figure 2: Inertial Measurement Unit (IMU)

```
Barometric Pressure and Altimeter Data
Temperature = 24.32 *C
Pressure = 779.70 hPa
Approx. Altitude = 2156.16 m
```

Figure 3: Barometric Pressure and Altimeter

```
LIDAR Data
Reading a measurement... Distance (mm): 956
Reading a measurement... out of range
Reading a measurement... Distance (mm): 8191
Reading a measurement... Distance (mm): 269
Reading a measurement... Distance (mm): 34
Reading a measurement... Distance (mm): 35
Reading a measurement... Distance (mm): 119
```

Figure 4: Time of Flight Micro-LIDAR Distance

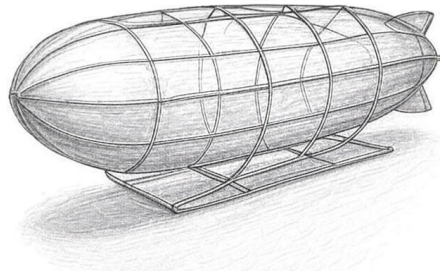
3 CUSTOMER REQUIREMENTS

The autonomous airship project is created to model how a large-scale version of this blimp would behave and function in the real world. The airship is designed to navigate autonomously in the real world and move from point A to point B, while simulating cargo transport. The requirements listed below are for the customer, as they provide value to their investment in the project. These requirements validate customer needs and provide a robust base with which our customers can start to create a real-world version and show the operational capabilities from the end user's perspective.

3.1 DURABILITY

3.1.1 DESCRIPTION

The airship will be designed to be reasonably durable and last in the outside environment. Since the large scale model has to face real-world conditions, the smaller model must be able to face proportional challenges as would a larger version.



3.1.2 SOURCE

CSE Senior Design project specifications

3.1.3 CONSTRAINTS

Durable enough to handle regular use and outside conditions.

3.1.4 STANDARDS

14 CFR Part 21.17(b) "Special Class" Airworthiness Criteria [4]
ISO 26262 [11]

3.1.5 PRIORITY

Moderate - The requirement is an excellent addition but is not critical to the main functions of the airship.

3.2 AERIAL REGULATION COMPLIANCE

3.2.1 DESCRIPTION

The airship must meet aerial regulations that a drone or a larger airship model would need to meet. This requirement allows the autonomous airship to be used and tested by the customer without having to worry about the regulations regarding the airship.

3.2.2 SOURCE

CSE Senior Design project specifications

3.2.3 CONSTRAINTS

Must meet regulations regarding drones and aerial electronic equipment to be used in public and private places. Must try to meet the local regulations regarding the safe and lawful operation of these devices.

3.2.4 STANDARDS

14 CFR Part 107 - Small Unmanned Aircraft Systems (sUAS) [3]

Texas Government Code Chapter 423 - Use of Unmanned Aircraft [1]

3.2.5 PRIORITY

Moderate - The requirement is an excellent addition but is not critical to the main functions of the airship, this would allow our customers to display the functions and operations of our airship in public, but the inclusion of this requirement is not critical to the functions of the airship.

3.3 VERTICAL TAKE OFF AND LANDING

3.3.1 DESCRIPTION

The airship must be able to vertically take off from the ground and land safely. It must perform these functions automatically with little input from the user other than the altitude to fly at and the route to take to the destination.

3.3.2 SOURCE

CSE Senior Design project specifications

3.3.3 CONSTRAINTS

The airship must take off vertically at the start of the route and then safely land at the end. The blimp must perform actions to stay in the air after take off. The frame of the blimp must be build accordingly to accommodate this feature.

3.3.4 STANDARDS

14 CFR Part 21 [4]

14 CFR Part 107 [3]

3.3.5 PRIORITY

HIGH - This is one of the main features that makes the airship autonomous.

3.4 CARGO TRANSPORTATION CAPABILITIES

3.4.1 DESCRIPTION

The airship is a small-scale version of larger airships that would be used to carry cargo. The airship should have a place to store cargo and for the blimp to accommodate the extra weight provided by the cargo.

3.4.2 SOURCE

CSE Senior Design project specifications

3.4.3 CONSTRAINTS

A large cargo hold able to store weights representing cargo. The airship must account for the extra weight when performing altitude operations.

3.4.4 STANDARDS

14 CFR Part 107 [3]

ICAO TI - Doc 9284 [10]

3.4.5 PRIORITY

LOW - The main purpose of the airship is to show the autonomous and navigational capabilities of the project. Transportation capabilities would elevate the project's appeal but are not critical to the main functions of the project.

3.5 MOTORS AND PROPELLERS TO CHANGE ALTITUDE

3.5.1 DESCRIPTION

Motors and propellers will be used to move the airship upwards and forwards. The upward propellers will lift the airship and adjust the altitude and the forward propellers will help move and rotate the airship while in the air. The speed of the motors will determine the speed of the airship's movements and its ability to rotate.

3.5.2 SOURCE

CSE Senior Design project specifications

3.5.3 CONSTRAINTS

Motors and propellers move the airship. The motor's speed and direction will need to be controlled in addition to the sequence of motors used.

3.5.4 STANDARDS

14 CFR Part 107 [3]

ICAO TI - Doc 9284 [10]

3.5.5 PRIORITY

HIGH

3.6 AUTO NAVIGATION

3.6.1 DESCRIPTION

The blimp will be able to autonomously navigate to a destination set by the user. The airship should be able to takeoff, adjusting its altitude, turn to face the correct direction, and travel to the destination without manual control. The navigation system should be able to use GPS coordinates and sensor data to detect and avoid simple obstacles while moving to the destination.

3.6.2 SOURCE

CSE Senior Design project specifications

3.6.3 CONSTRAINTS

The blimp should autonomously complete all stages of flight, including takeoff, travel, and landing. The blimp should use GPS to define routes and sensor data to maintain the route and respond to any obstacles or disturbances. Communication with the ground station should be present throughout the flight to send telemetry data.

3.6.4 STANDARDS

14 CFR Part 107 [3]

ICAO TI - Doc 9284 [10]

3.6.5 PRIORITY

HIGH

4 PACKAGING REQUIREMENTS

This section details how the project will be packaged to the customer in the end. The project will include hardware and software components. The restrictions here will describe how the parts need to be packaged, so that the customer can safely and securely receive the parts, and also describe items that the customer will have to supply themselves, such as batteries.

4.1 HARDWARE COMPONENTS

4.1.1 DESCRIPTION

The three hardware part consists of the gondola, the rigid frame, and the base.

4.1.2 SOURCE

CSE Senior Design project specifications

4.1.3 CONSTRAINTS

The parts are split into individual packages and no batteries will be provided. Packaging will follow normal shipping procedures and postal service standards.

4.1.4 STANDARDS

IATA Dangerous Goods Regulations - DGR [8]

4.1.5 PRIORITY

Low

4.2 SOFTWARE COMPONENTS

4.2.1 DESCRIPTION

The software components include the Arduino, the Raspberry pi, ground station (laptop or computer), and the GitHub where the code will be stored.

4.2.2 SOURCE

CSE Senior Design project specifications

4.2.3 CONSTRAINTS

The code for the devices need to be organized, documented, and securely stored on the GitHub to be shared with the customer. Access needs to be limited to only the developers and customers.

4.2.4 STANDARDS

GitHub Docs: Best practices for repositories [6]

GitHub Docs: About security advisors / Recommended security practices [5]

4.2.5 PRIORITY

Moderate

5 PERFORMANCE REQUIREMENTS

The requirements listed below list the abilities that the airship must contain. These requirements list the items that are very critical to the operations of the blimp and are fundamental to the basic operations of our project.

5.1 OBSTACLE AVOIDANCE DURING FLIGHT

5.1.1 DESCRIPTION

The system shall detect and avoid obstacles in its flight path using sensor input.

5.1.2 SOURCE

Project charter

5.1.3 CONSTRAINTS

Obstacle detection systems are constrained by sensor resolution, range, and latency. Weather conditions and reflective surfaces can disrupt sensor reliability. Constraints include safety, environmental interference, and system maintainability.

5.1.4 STANDARDS

IEEE 1937: Drone Payloads

5.1.5 PRIORITY

High: Required for continuous and safe autonomous flight.

5.2 BUOYANCY GENERATION

5.2.1 DESCRIPTION

The airship will use helium inside its ballonnet and envelope to help generate lift.

5.2.2 SOURCE

Goodyear's Pilgrim Blimp [2]

5.2.3 CONSTRAINTS

The use of helium has a plethora of constraints such as usability, maintainability, sustainability, and cost. Although helium is the second most abundant element in the universe, it is a scarce resource globally. Moreover, helium has a wide range of critical uses, including medical applications, manufacturing, and research. These facts contribute to the high cost of the element which in turn affects the sustainability, maintainability, and usability if the product is in mass production.

5.2.4 STANDARDS

OSHA Compliance: Part 1910 - Occupational Safety and Health Standards

5.2.5 PRIORITY

Critical

5.3 5 - 10 MINUTE FLIGHT TIME

5.3.1 DESCRIPTION

The airship should be able to navigate routes that take 5 to 10 minutes. This would suffice to prove the functions of the airship while providing an indication of the power to size ratio.

5.3.2 SOURCE

CSE Senior Design project specifications

5.3.3 CONSTRAINTS

The length of the routes taken by the blimp are limited by the power provided by the battery. The larger the battery, the longer the routes that will become navigable. The power draw of electrical components, such as motors, micro controllers, breakout boards, and sensors will increase the need for a larger battery.

5.3.4 STANDARDS

14 CFR Part 107 [3]

5.3.5 PRIORITY

Moderate - A good flight time is a very good indication for the viability of creating larger models of our autonomous airship. Although this is necessary for a small demo blimp, meant to represent the capabilities of a large scale version, it is a good feature to have a longer flight time. This better showcase the navigational abilities of the blimp on longer flights.

5.4 PID CONTROLLED ALTITUDE ADJUSTMENT

5.4.1 DESCRIPTION

The airship should be able to change its altitude and fly at the set altitude by the controller. The altitude adjust must be controlled by a PID controller and must accommodate for environmental factors to keep the airship at a set altitude.

5.4.2 SOURCE

CSE Senior Design project specifications

5.4.3 CONSTRAINTS

The airship must maintain a set altitude regardless of additional weight and other environmental factors.

5.4.4 STANDARDS

14 CFR Part 107 [3]

5.4.5 PRIORITY

HIGH - This requirement prioritizes one of the main navigation abilities of the airship. The airship must accommodate for outside factors and maintain set altitude.

5.5 WIRELESS COMMUNICATION WITH GROUND STATION

5.5.1 DESCRIPTION

The airship should be able to wirelessly communicate with a ground station to receive and control commands and send telemetry data. The connection should allow real-time adjustments and monitoring.

5.5.2 SOURCE

CSE Senior Design project specifications

5.5.3 CONSTRAINTS

The system is constrained by range, bandwidth and reliability of Wi-fi. There may be interference from nearby systems or environmental factors.

5.5.4 STANDARDS

IEEE 802.11 - Wireless Local Area Network (Wi-Fi) [9]

5.5.5 PRIORITY

HIGH - Wireless communication is critical for controlling and monitoring the airship during flight.

6 SAFETY REQUIREMENTS

The safety requirements section for the autonomous cargo airship lays out the principles implemented to ascertain safe operation of the blimp. These principles apply to testing, manufacturing, and consumer use of the product.

6.1 LABORATORY EQUIPMENT LOCKOUT/TAGOUT (LOTO) PROCEDURES

6.1.1 DESCRIPTION

Any fabrication equipment provided used in the development of the project shall be used in accordance with OSHA standard LOTO procedures. Locks and tags are installed on all equipment items that present use hazards, and ONLY the course instructor or designated teaching assistants may remove a lock. All locks will be immediately replaced once the equipment is no longer in use.

6.1.2 SOURCE

CSE Senior Design laboratory policy

6.1.3 CONSTRAINTS

Equipment usage, due to lock removal policies, will be limited to availability of the course instructor and designed teaching assistants.

6.1.4 STANDARDS

Occupational Safety and Health Standards 1910.147 - The control of hazardous energy (lockout/tagout).

6.1.5 PRIORITY

Critical

6.2 NATIONAL ELECTRIC CODE (NEC) WIRING COMPLIANCE

6.2.1 DESCRIPTION

Any electrical wiring must be completed in compliance with all requirements specified in the National Electric Code. This includes wire runs, insulation, grounding, enclosures, over-current protection, and all other specifications.

6.2.2 SOURCE

CSE Senior Design laboratory policy

6.2.3 CONSTRAINTS

High voltage power sources, as defined in NFPA 70, will be avoided as much as possible in order to minimize potential hazards.

6.2.4 STANDARDS

NFPA 70

6.2.5 PRIORITY

Critical

6.3 RIA ROBOTIC MANIPULATOR SAFETY STANDARDS

6.3.1 DESCRIPTION

Robotic manipulators, if used, will either housed in a compliant lockout cell with all required safety interlocks, or certified as a "collaborative" unit from the manufacturer.

6.3.2 SOURCE

CSE Senior Design laboratory policy

6.3.3 CONSTRAINTS

Collaborative robotic manipulators will be preferred over non-collaborative units in order to minimize potential hazards. Sourcing and use of any required safety interlock mechanisms will be the responsibility of the engineering team.

6.3.4 STANDARDS

ANSI/RIA R15.06-2012 American National Standard for Industrial Robots and Robot Systems, RIA TR15.606-2016 Collaborative Robots

6.3.5 PRIORITY

Critical

6.4 FAIL-SAFE CONTINGENCIES

6.4.1 DESCRIPTION

The autonomous cargo airship will implement a fail-safe that will land the blimp from its takeoff point whenever it is disconnected from the ground station or low on battery. [14]

6.4.2 SOURCE

EU Drone Port

6.4.3 CONSTRAINTS

The fail-safe mechanism must be interoperable between the IMU and microcontroller once the mode is enabled, allowing the blimp to guide itself back to its origin. Moreover, once the fail-safe is engaged, the blimp will follow safety and welfare according to Federal Aviation Administration (FAA) regulations.

6.4.4 STANDARDS

Code of Federal Regulations: FAA - Small Unmanned Aircraft Systems (UAS) Regulations (Part 107) [3]

6.4.5 PRIORITY

High

6.5 VISION-BASED LANDING ZONE DETECTION

6.5.1 DESCRIPTION

The system shall use an onboard camera to analyze terrain and detect safe, flat surfaces for landing during autonomous operation.

6.5.2 SOURCE

Project charter

6.5.3 CONSTRAINTS

Camera vision can be affected by environmental factors such as lighting, shadows, fog, or weather. The design must account for environmental safety, usability, and maintainability constraints.

6.5.4 STANDARDS

ISO 21448, ISO 12640

6.5.5 PRIORITY

High: Critical for safe autonomous landings in changing environments.

7 MAINTENANCE & SUPPORT REQUIREMENTS

The maintenance and support for the autonomous cargo airship will be outlined in the troubleshooting manual. Helium management will be done by the consumer with the help of the manual. The consumer must adhere to OSHA regulations on compressed helium handling to ensure safety.

7.1 HELIUM MANAGEMENT

7.1.1 DESCRIPTION

The consumer will be able to load and unload helium gas inside the ballonnet and envelope through the fill spout. The instructions on how to execute the procedure will be detailed in the troubleshooting manual.

7.1.2 SOURCE

Project Airship

7.1.3 CONSTRAINTS

Usability: Depending on where the consumer is located, refilling the ballonnet may be cumbersome if helium gas tanks are not readily available.

Cost: The cost of helium will vary depending on the current economic climate.

Sustainability: Refilling of helium should be at a minimal depending on the condition of the envelope. If the envelope is in great condition, the amount of times to refill the ballonnet will be at a minimum.

Maintainability: Loading and unloading the ballonnet with helium will be consumer friendly due to how the fill spout will be designed.

7.1.4 STANDARDS

OSHA Compliance: Part 1910 - Occupational Safety and Health Standards

7.1.5 PRIORITY

Critical

8 SECURITY REQUIREMENTS

This section describes the requirements that must be implemented for secured operations and secure coding practices used to program the software used in this project.

8.1 SECURE USER GUIDE

8.1.1 DESCRIPTION

The project will provide a user guide that explains the purpose of using the Arduino and Raspberry Pi. Using different devices that do not support the provided code will cause complications and harm the airship.

8.1.2 SOURCE

CSE Senior Design project specifications

8.1.3 CONSTRAINTS

Creating a user guide to help users to securely use the product

8.1.4 STANDARDS

ISO/IEC/IEEE 82079-1:2019 [13]

8.1.5 PRIORITY

High

8.2 SECURE CODING PRACTICES

8.2.1 DESCRIPTION

The software for the airship must be written with secure coding practices following the standards listed below.

8.2.2 SOURCE

CSE Senior Design project specifications

8.2.3 CONSTRAINTS

Secure coding practices will need to be used for C++ (Arduino Language) and for Python (Ground Station and Raspberry Pi). Code will be documented and stored securely.

8.2.4 STANDARDS

ISO 14882 C++ Programming Language [12]

Python Language Reference [15]

8.2.5 PRIORITY

High

9 OTHER REQUIREMENTS

The requirement listed below detail things that are not consistent with other sections. These include details that are convenience features that make the user experience better. For example, the map API integration shows the user of the airship's path while it is airborne.

9.1 MAPS API INTEGRATION

9.1.1 DESCRIPTION

The system shall use a Maps API to define waypoints and plan autonomous routes from takeoff to landing. This will allow for flexible navigation based on geographic input.

9.1.2 SOURCE

Project charter

9.1.3 CONSTRAINTS

This feature requires a stable internet connection for accessing map data and may be impacted by API usage limits, availability, and licensing. Additionally, the system must confirm that GPS signal strength is sufficient in the deployment environment. These restrictions relate to usability, interoperability, and environmental design constraints

9.1.4 STANDARDS

ISO 19101 through 19159:Geographic Information Systems, ISO 17185: Journey Planning Systems

9.1.5 PRIORITY

Medium: Improves mission planning but is not critical to core flight operations.

9.2 WI-FI BASED COMMUNICATION LINK

9.2.1 DESCRIPTION

The system shall use Wi-Fi to wirelessly communicate between the ground station and airship. The link will be used to send control commands to the blimp and send telemetry data to the ground station.

9.2.2 SOURCE

CSE Senior Design project specifications

9.2.3 CONSTRAINTS

This requirement relies on the strength and reliability of the Wi-Fi signal. Factors such as distance and environmental obstacles will affect performance.

9.2.4 STANDARDS

IEEE 802.11 - Wireless Local Area Network (Wi-Fi) [9]

9.2.5 PRIORITY

High: Important for system usability and autonomous actions of the airship.

10 FUTURE ITEMS

In this last section, we describe the requirements we were not able to include due to time, budget, and feasibility constraints. These are requirements we discussed and wanted to include if advanced progress was made on the project milestones and if enough resources were present. Although these requirements are not critical to the function of the airship, they are excellent additions to the capabilities of the project. Adding these requirements to future versions of this project would elevate the project to new heights and add to the marketability of the product.

10.1 CARGO TRANSPORTATION CAPABILITIES

10.1.1 DESCRIPTION

The airship is a small-scale version of larger airships that would be used to carry cargo. The airship should have a place to store cargo and for the blimp to accommodate the extra weight provided by the cargo.

10.1.2 SOURCE

CSE Senior Design project specifications

10.1.3 CONSTRAINTS

A large cargo hold able to store weights representing cargo. The airship must account for the extra weight when performing altitude operations.

10.1.4 STANDARDS

14 CFR Part 107 [3]

ICAO TI - Doc 9284 [10]

10.1.5 PRIORITY

LOW - The main purpose of the airship is to show the autonomous and navigational capabilities of the project. Transportation capabilities would elevate the project's appeal but are not critical to the main functions of the project.

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